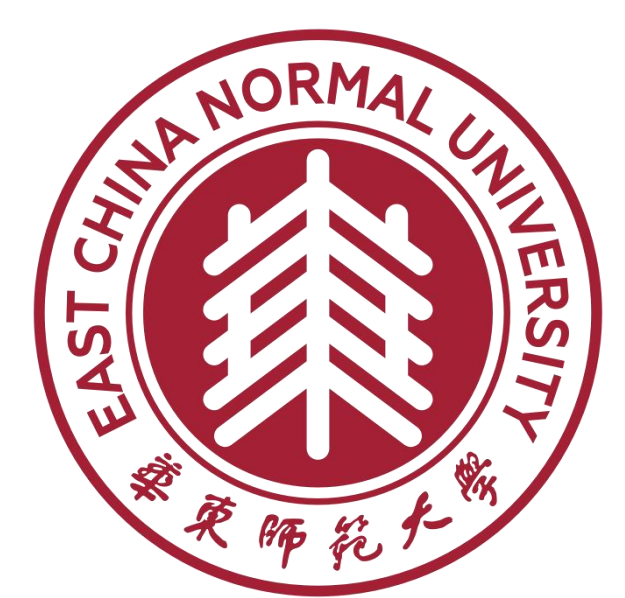


MoRI: Learning Motivation-Grounded Reasoning for Scientific Ideation in Large Language Models



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MoRI internalizes scientific ideation by learning motivation-grounded reasoning, yield ideas that are more novel, technically rigorous, and feasible than strong LLM and agentic baselines.

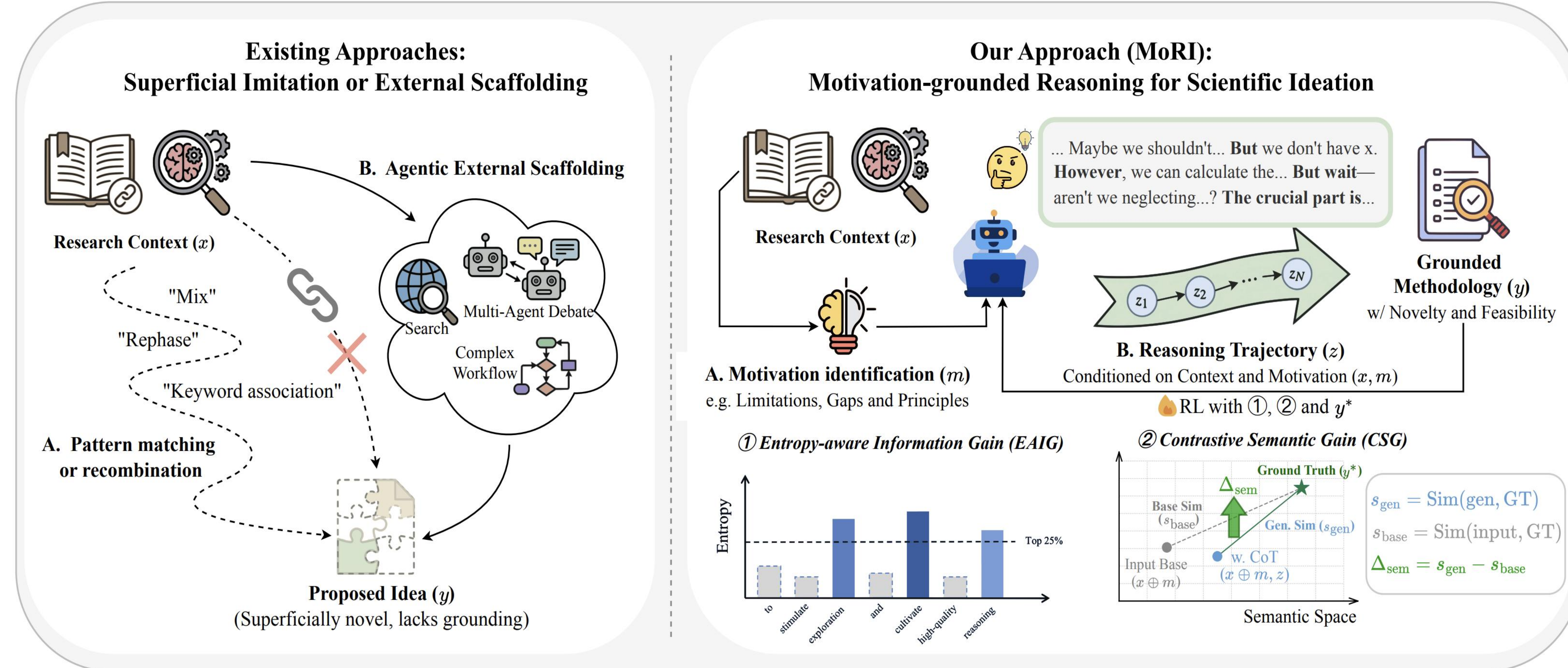
Problem & Motivation

- Scientific ideation requires mapping a research context to a feasible methodology through principled motivation.
- Existing LLM approaches often rely on surface-level recombination or external agentic scaffolding.
- These methods can appear novel but often lack technical depth and scientific grounding.
- We instead train the model to reason from context $\rightarrow$ motivation $\rightarrow$ methodology.

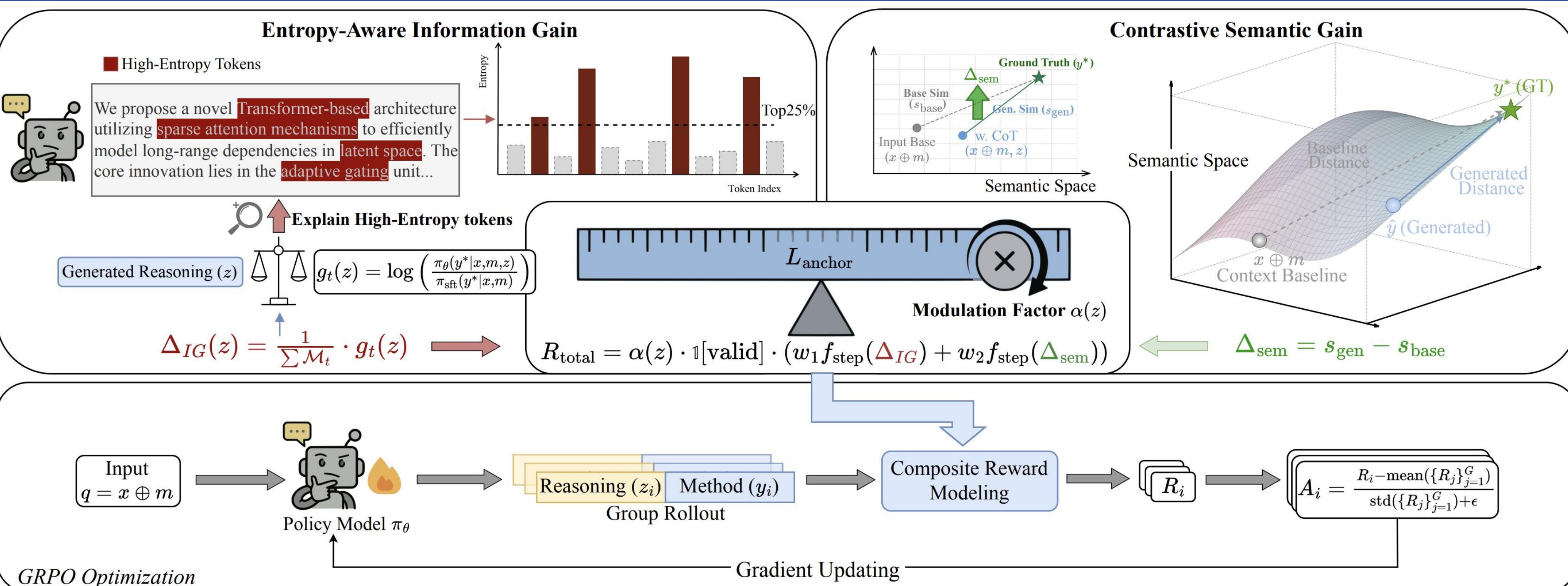
Contributions

- Formulate scientific ideation as motivation-grounded reasoning with intermediate motivation and reasoning trajectory.
- Propose MoRI, combining supervised initialization and reinforcement learning to internalize scientific reasoning.
- Design composite rewards that balance technical depth and conceptual alignment.

Conceptual Comparison



Method & Reward Design



Experiments & Results

Model	ICLR Test Set					NeurIPS OOD Test Set					Overall				
	Nov.	Rig.	Feas.	Mean	Δ	Nov.	Rig.	Feas.	Mean	Δ	Nov.	Rig.	Feas.	Mean	Δ
Commercial Models															
GPT-4o	2.51	2.78	2.79	2.69	+18.6%	2.57	2.87	2.88	2.77	+13.7%	2.54	2.83	2.84	2.74	+16.1%
Claude-3.5-Sonnet	3.39	3.07	2.82	3.09	+3.2%	3.42	3.03	2.94	3.13	+0.6%	3.40	3.05	2.89	3.11	+2.3%
Agentic Frameworks															
AI-Scientist-V2 [†] (Sonnet)	3.45	3.08	2.89	3.14	+1.6%	—	—	—	—	—	—	—	—	—	—
AI-Scientist-V2 (Haiku)	2.74	2.46	2.89	2.70	+18.1%	—	—	—	—	—	—	—	—	—	—
AI-Scientist-V2 (GPT-4o)	2.48	2.67	2.98	2.71	+17.7%	2.28	2.55	2.61	2.48	+27.0%	2.38	2.60	2.82	2.60	+22.3%
ResearchAgent	2.66	2.51	2.63	2.60	+22.7%	2.58	2.22	2.32	2.37	+32.9%	2.63	2.39	2.50	2.50	+27.2%
VirSci	2.21	2.26	2.28	2.25	+41.8%	2.16	2.27	2.27	2.23	+41.3%	2.19	2.26	2.27	2.24	+42.0%
Ours															
Full-SFT	3.10	2.87	3.02	2.99	+6.7%	2.85	2.76	2.94	2.85	+10.5%	2.99	2.82	2.99	2.93	+8.5%
MoRI	3.31	3.16	3.11	3.19	—	3.30	3.00	3.16	3.15	—	3.31	3.09	3.13	3.18	—
Ground Truth	3.63	3.69	3.44	3.58	Ref.	3.51	3.69	3.57	3.59	Ref.	3.58	3.69	3.49	3.59	Ref.

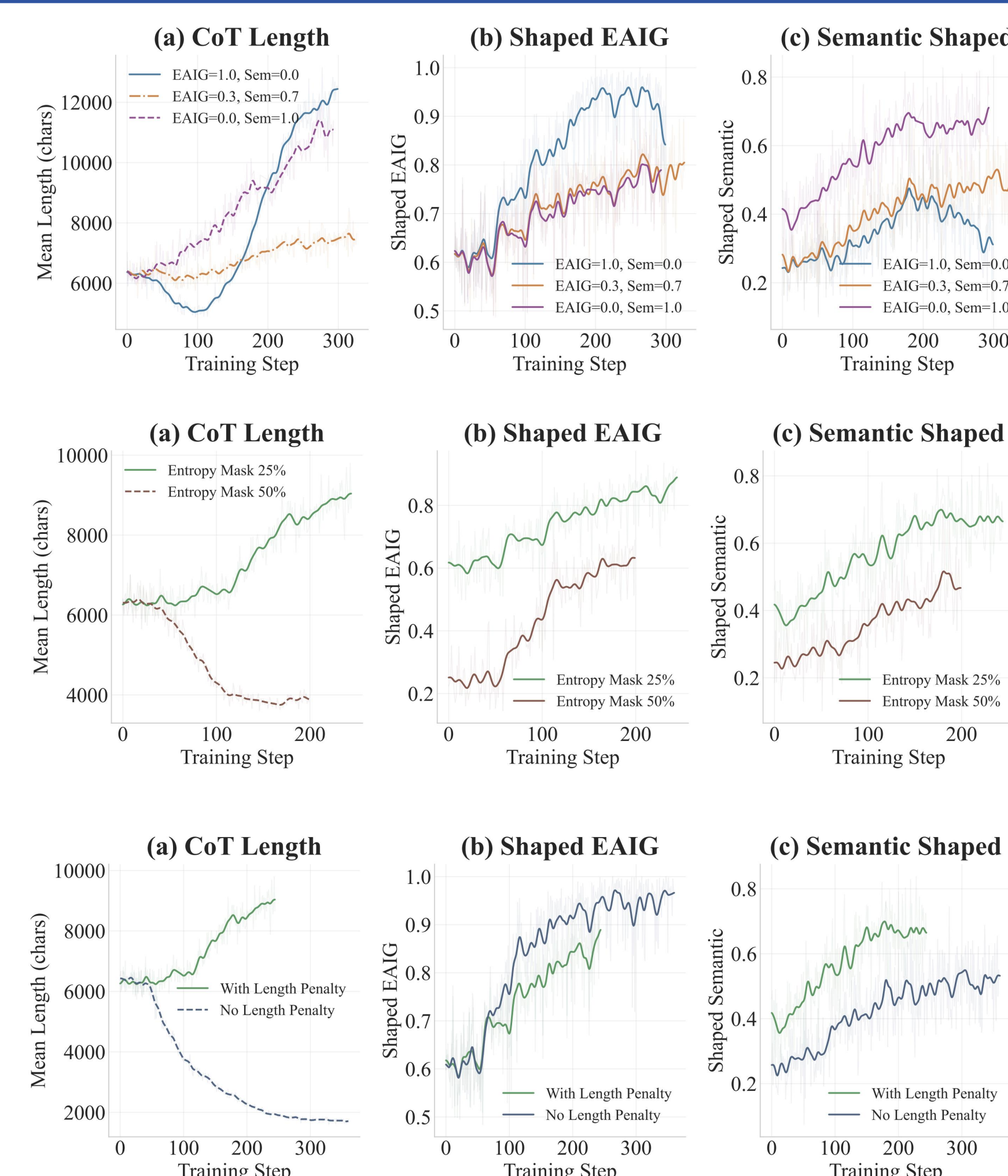
MoRI achieves the best overall model performance: 3.18, outperforming Full-SFT(+8.5%) and Claude-3.5-Sonnet(+2.3%), while improving rigor and feasibility.

- Test sets:** ICLR in-domain test set with 83 papers; NeurIPS out-of-domain test set with 67 papers; combined 150 papers.
- Metrics:** Novelty, Technical Rigor, Feasibility.
- Evaluation:** Retrieval-augmented LLM judge validated against human experts.
- Human-LLM alignment:** Pearson r: Overall 0.715, Novelty 0.723, Rigor 0.751, Feasibility 0.682.

Ablation Study & Dynamics

Ablation	Setting	Nov.	Rig.	Feas.	Mean	Δ	Note
Motivation Conditioning	Full-SFT Direct ($x \rightarrow y$)	2.80	2.69	2.81	2.75	—	Direct generation
	Full-SFT Two-stage ($x \rightarrow m \rightarrow y$)	2.85	2.76	2.94	2.85	—	Motivation-guided
	MoRI	3.30	3.00	3.16	3.15	+0.30	RL on two-stage
Reward Composition	EAIG only ($w_s = 0.0, w_e = 1.0$)	2.68	2.22	2.63	2.51	—	Technical-detail reward only
	CSG only ($w_s = 1.0, w_e = 0.0$)	3.16	2.93	3.06	3.05	—	Semantic alignment only
	Balanced ($w_s = 0.5, w_e = 0.5$)	3.32	2.96	2.98	3.09	—	Equal weights
	Optimal ($w_s = 0.7, w_e = 0.3$)	3.34	3.04	3.07	3.15	—	Best reward mix
Entropy Filtering	Top 50% entropy mask	3.16	2.78	3.00	2.98	—	Looser filter
	Top 25% entropy mask	3.32	2.96	2.98	3.09	+3.7%	Cleaner technical signal
Length Anchoring	$w_s = 0.7, w_e = 0.3$, w/o LA	3.22	2.94	3.00	3.05	—	No length control
	$w_s = 0.7, w_e = 0.3$, w/ LA	3.34	3.04	3.07	3.15	+3.2%	Stabilized reasoning
	$w_s = 0.5, w_e = 0.5$, w/o LA	2.96	2.76	2.95	2.89	—	No length control
	$w_s = 0.5, w_e = 0.5$, w/ LA	3.32	2.96	2.98	3.09	+6.9%	Stabilized reasoning

- Motivation conditioning** improves over direct generation, validating the context $\rightarrow$ motivation $\rightarrow$ methodology formulation.
- Combining CSG and EAIG** yields stronger results than either reward alone, showing the need to balance semantic direction and technical depth.
- Top-25% entropy filtering and length anchoring** stabilize optimization, preventing noisy reward signals and shallow reasoning shortcuts.



Resources



Paper



Code